

Inconsistent type returned by <class 'pennylane.vqe.vqe.Hamiltonian'> pr

<https://github.com/PennyLaneAI/pennylane/issues/1265>

ROW 83

Inconsistent type returned by <class 'pennylane.vqe.vqe.Hamiltonian'> property ops #1265

New issue

Closed



aemccoy opened on May 1, 2021

Issue description

When `vqe.Hamiltonian` contains terms consisting of a single qubit operator, e.g., `H=2*Z0`, `H.ops` returns a list of operators whose type depends on whether `Hamiltonian` is constructed directly from pennylane pauli operator `2*qml.PauliZ(0)` or if `Hamiltonian` is converted to `vqe.Hamiltonian` from `openfermion QubitOperator` using `qml.qchem.convert_observable`.

- **Expected behavior:** Properties of `Hamiltonian` consistently return same type for the same operator independent of construction
- **Actual behavior:** See below
- **Reproduces how often:** Always

- **System information:** (post the output of `import pennylane as qml; qml.about()`)
Name: PennyLane
Version: 0.16.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: <https://github.com/XanaduAI/pennylane>
Author: None
Author-email: None
License: Apache License 2.0
Location: /opt/anaconda3/envs/pennylane/lib/python3.9/site-packages
Requires: toml, semantic-version, networkx, autograd, appdirs, numpy, scipy
Required-by: PennyLane-qiskit, PennyLane-Qchem
Platform info: macOS-10.16-x86_64-i386-64bit
Python version: 3.9.4
Numpy version: 1.20.1
Scipy version: 1.6.1
Installed devices:

- default.gaussian (PennyLane-0.16.0.dev0)
- default.mixed (PennyLane-0.16.0.dev0)
- default.qubit (PennyLane-0.16.0.dev0)
- default.qubit.autograd (PennyLane-0.16.0.dev0)
- default.qubit.jax (PennyLane-0.16.0.dev0)
- default.qubit.tf (PennyLane-0.16.0.dev0)
- default.tensor (PennyLane-0.16.0.dev0)
- default.tensor.tf (PennyLane-0.16.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.15.0)
- qiskit.basicaer (PennyLane-qiskit-0.15.0)
- qiskit.ibmq (PennyLane-qiskit-0.15.0)

Source code and tracebacks

If the `vqe.Hamiltonian` is constructed directly in pennylane

```
A=2*qml.PauliZ(0)
print("A: ", type(A))
print(A)
print("A.ops: ", type(A.ops))
print(A.ops)
print("A.ops[0]: ", type(A.ops[0]))
print(A.ops[0])
print("A.ops[0].name: ", type(A.ops[0].name))
print(A.ops[0].name)
```

Assignees

agran2018

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



Returns

```
A: <class 'pennylane.vqe.vqe.Hamiltonian'>
(2) [Z0]
A.ops: <class 'list'>
[PauliZ(wires=[0])]
A.ops[0]: <class 'pennylane.ops.qubit.PauliZ'>
PauliZ(wires=[0])
A.ops[0].name: <class 'str'>
PauliZ
```

If equivalent vqe.Hamiltonian is obtained from openfermion QubitOperator

```
qubit_op=QubitOperator("Z0",2)
B=qml.qchem.convert_observable(qubit_op)

print("B: ", type(B))
print(B)
print("B.ops: ",type(B.ops))
print(B.ops)
print("B.ops[0]: ",type(B.ops[0]))
print(B.ops[0])
print("B.ops[0].name: ",type(B.ops[0].name))
print(B.ops[0].name)
```

return

```
B: <class 'pennylane.vqe.vqe.Hamiltonian'>
(2) [Z0]
B.ops: <class 'list'>
[PauliZ(wires=[0])]
B.ops[0]: <class 'pennylane.operation.Tensor'>
PauliZ(wires=[0])
B.ops[0].name: <class 'list'>
['PauliZ']
```



josh146 on May 3, 2021

Member ...

Thanks for noting this @aemccoy! This is definitely something we can fix to make more consistent.



josh146 assigned agran2018 on May 3, 2021



agran2018 mentioned this in 2 pull requests on Aug 13, 2021

Fix github issue #1265 #1523

Fix GitHub issue #1265 #1525



agran2018 added a commit that references this issue on Aug 13, 2021

Fix GitHub issue #1265 (#1525) ...

Verified 6e5cd0e



agran2018 closed this as **completed** on Aug 13, 2021



josh146 added a commit that references this issue on Aug 17, 2021

Merge `v0.17.0-rc0` into `master` (#1540) ...

Verified 62f7c10

Fix GitHub issue #1265 #1525

<> Code

Merged agran2018 merged 4 commits into `v0.17.0-rc0` from `fix-github-issue-1265` on Aug 13, 2021

Conversation 3 Commits 4 Checks 0 Files changed 3 +38 -4



agran2018 commented on Aug 13, 2021

Contributor

Context: Fix github issue #1265. This PR ensures the type consistency of the qubit Hamiltonian terms constructed by the 'convert_observable' function from an OpenFermion QubitOperator with respect to the same observable built directly using PennyLane operations.

Description of the Change:

The fix was implemented in the `_qubit_operator_to_terms` function of the `qchem/structure` module. An extra unit test was added to `test_convert_observable.py`

Benefits:

Properties of Hamiltonian consistently return same type.

Possible Drawbacks:

None

Related GitHub Issues:

[#1265](#)



Reviewers

rmoyard ✓

antalszava ●

Assignees

agran2018

zeyueN

Labels

[review-ready](#)

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

Notifications

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3 participants



agran2018 added 3 commits 4 years ago

Modify '_qubit_operator_to_terms' and '_terms_to_qubit_operator' func...

a9b0ba7

Add new testing function to 'test_convert_observable.py'

a72ddfe

Add the bug fix item to CHANGELOG.md

719c14d

agran2018 added the [review-ready](#) label on Aug 13, 2021

agran2018 requested review from antalszava and rmoyard 4 years ago

agran2018 assigned agran2018 and zeyueN on Aug 13, 2021



github-actions bot commented on Aug 13, 2021

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [github/CHANGELOG.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Update CHANGELOG.md

✓ 5867e14